

# STA323 Assignment3

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## Q1

(1)

本小问代码为Q1\_1.py，我另外写了Q1\_1\_test1/2.py代码用于检验输出结果，检验程序的运行截图如下：

| window_start        | window_end          | device_type | cn                               | count |
|---------------------|---------------------|-------------|----------------------------------|-------|
| 2016-03-20 11:20:54 | 2016-03-20 11:20:55 | therm-stick | Republic of Korea                | 23    |
| 2016-03-20 11:20:55 | 2016-03-20 11:20:56 | meter-gauge | New Zealand                      | 10    |
| 2016-03-20 11:20:54 | 2016-03-20 11:20:55 | meter-gauge | Bosnia and Herzegovina           | 1     |
| 2016-03-20 11:20:54 | 2016-03-20 11:20:55 | sensor-pad  | Saint Vincent and the Grenadines | 1     |
| 2016-03-20 11:20:55 | 2016-03-20 11:20:56 | therm-stick | Greece                           | 2     |
| 2016-03-20 11:20:54 | 2016-03-20 11:20:55 | sensor-pad  | Finland                          | 8     |
| 2016-03-20 11:20:54 | 2016-03-20 11:20:55 | device-mac  | Finland                          | 5     |
| 2016-03-20 11:20:54 | 2016-03-20 11:20:55 | device-mac  | Cyprus                           | 1     |
| 2016-03-20 11:20:54 | 2016-03-20 11:20:55 | meter-gauge | Bulgaria                         | 9     |
| 2016-03-20 11:20:55 | 2016-03-20 11:20:56 | therm-stick | Belarus                          | 1     |
| 2016-03-20 11:20:54 | 2016-03-20 11:20:55 | sensor-pad  | Republic of Lithuania            | 4     |
| 2016-03-20 11:20:55 | 2016-03-20 11:20:56 | meter-gauge | Chile                            | 7     |
| 2016-03-20 11:20:55 | 2016-03-20 11:20:56 | therm-stick | Puerto Rico                      | 2     |
| 2016-03-20 11:20:55 | 2016-03-20 11:20:56 | therm-stick | Ecuador                          | 1     |
| 2016-03-20 11:20:55 | 2016-03-20 11:20:56 | sensor-pad  | Puerto Rico                      | 6     |
| 2016-03-20 11:20:54 | 2016-03-20 11:20:55 | therm-stick | Australia                        | 8     |
| 2016-03-20 11:20:54 | 2016-03-20 11:20:55 | meter-gauge | Greece                           | 4     |
| 2016-03-20 11:20:55 | 2016-03-20 11:20:56 | sensor-pad  | United Arab Emirates             | 4     |
| 2016-03-20 11:20:55 | 2016-03-20 11:20:56 | device-mac  | Puerto Rico                      | 1     |
| 2016-03-20 11:20:54 | 2016-03-20 11:20:55 | sensor-pad  | Greece                           | 7     |

```
only showing top 20 rows
row count = 709
2016-03-20 11:20:55 meter-gauge New Zealand Spain
=== Result Row Count ===
709
max_event_time = 2016-03-20 11:21:01
=== Expected Closed Row Count ===
709
expected_closed - result = 0
result - expected_closed = 0
```

(2)

本小问代码为Q1\_2.py，我同样写了Q1\_2\_test.py代码用于检验，结果如下：

| window_start        | window_end          | cn                               | low_battery_device_count |
|---------------------|---------------------|----------------------------------|--------------------------|
| 2016-03-20 11:20:55 | 2016-03-20 11:20:56 | Hashemite Kingdom of Jordan      | 1                        |
| 2016-03-20 11:20:55 | 2016-03-20 11:20:56 | Bosnia and Herzegovina           | 2                        |
| 2016-03-20 11:20:55 | 2016-03-20 11:20:56 | Saint Kitts and Nevis            | 1                        |
| 2016-03-20 11:20:55 | 2016-03-20 11:20:56 | United Arab Emirates             | 5                        |
| 2016-03-20 11:20:55 | 2016-03-20 11:20:56 | Norway                           | 46                       |
| 2016-03-20 11:20:54 | 2016-03-20 11:20:55 | Dominican Republic               | 2                        |
| 2016-03-20 11:20:55 | 2016-03-20 11:20:56 | Portugal                         | 26                       |
| 2016-03-20 11:20:55 | 2016-03-20 11:20:56 | Saint Vincent and the Grenadines | 1                        |
| 2016-03-20 11:20:55 | 2016-03-20 11:20:56 | Morocco                          | 4                        |
| 2016-03-20 11:20:55 | 2016-03-20 11:20:56 | France                           | 180                      |
| 2016-03-20 11:20:55 | 2016-03-20 11:20:56 | Haiti                            | 1                        |
| 2016-03-20 11:20:55 | 2016-03-20 11:20:56 | Oman                             | 1                        |
| 2016-03-20 11:20:55 | 2016-03-20 11:20:56 | South Africa                     | 13                       |
| 2016-03-20 11:20:54 | 2016-03-20 11:20:55 | Philippines                      | 10                       |
| 2016-03-20 11:20:54 | 2016-03-20 11:20:55 | Syria                            | 1                        |
| 2016-03-20 11:20:54 | 2016-03-20 11:20:55 | United Arab Emirates             | 3                        |
| 2016-03-20 11:20:55 | 2016-03-20 11:20:56 | Republic of Korea                | 366                      |
| 2016-03-20 11:20:55 | 2016-03-20 11:20:56 | French Polynesia                 | 1                        |
| 2016-03-20 11:20:54 | 2016-03-20 11:20:55 | Hashemite Kingdom of Jordan      | 1                        |
| 2016-03-20 11:20:54 | 2016-03-20 11:20:55 | Cyprus                           | 6                        |
| 2016-03-20 11:20:55 | 2016-03-20 11:20:56 | Dominica                         | 1                        |
| 2016-03-20 11:20:55 | 2016-03-20 11:20:56 | Afghanistan                      | 1                        |
| 2016-03-20 11:20:55 | 2016-03-20 11:20:56 | Israel                           | 17                       |
| 2016-03-20 11:20:54 | 2016-03-20 11:20:55 | Kazakhstan                       | 1                        |
| 2016-03-20 11:20:54 | 2016-03-20 11:20:55 | Cayman Islands                   | 1                        |
| 2016-03-20 11:20:54 | 2016-03-20 11:20:55 | Hong Kong                        | 23                       |
| 2016-03-20 11:20:54 | 2016-03-20 11:20:55 | Taiwan                           | 29                       |
| 2016-03-20 11:20:54 | 2016-03-20 11:20:55 | Saint Vincent and the Grenadines | 0                        |
| 2016-03-20 11:20:55 | 2016-03-20 11:20:56 | Cayman Islands                   | 0                        |
| 2016-03-20 11:20:55 | 2016-03-20 11:20:56 | Czech Republic                   | 54                       |

only showing top 30 rows  
274

## Q2

### (1)

代码文件为Q2\_1.py，经过调试，我将时间戳单位除以了1e12,来得到正常的时间节点，运行结果如下：

| window_start        | window_end          | user | count |
|---------------------|---------------------|------|-------|
| 2015-02-22 08:36:00 | 2015-02-22 08:42:00 | a    | 2     |
| 2015-02-22 08:36:00 | 2015-02-22 08:42:00 | a    | 1     |
| 2015-02-22 08:39:00 | 2015-02-22 08:45:00 | a    | 2     |

### (2)

代码文件为Q2\_2.py，运行结果如下：

| window_start        | window_end          | user | count |
|---------------------|---------------------|------|-------|
| 2015-02-22 08:36:00 | 2015-02-22 08:42:00 | a    | 1     |
| 2015-02-22 08:36:00 | 2015-02-22 08:42:00 | a    | 2     |
| 2015-02-22 08:39:00 | 2015-02-22 08:45:00 | a    | 1     |

## Q3

本题前两问运行结果如下

```
Best validation RMSE: 42414.12386805411
26/05/08 15:00:51 WARN DAGScheduler: Broadcasting large task binary with size 1023.6 KiB
26/05/08 15:00:51 WARN DAGScheduler: Broadcasting large task binary with size 1024.3 KiB
26/05/08 15:00:51 WARN DAGScheduler: Broadcasting large task binary with size 1024.9 KiB
26/05/08 15:00:51 WARN DAGScheduler: Broadcasting large task binary with size 1026.1 KiB
26/05/08 15:00:51 WARN DAGScheduler: Broadcasting large task binary with size 1028.4 KiB
Prediction finished.
Output directory: ./q3_dt_predictions_output
Please rename the generated part-*.csv to predictions.csv if needed.
26/05/08 22:46:56 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Prediction finished.
Output directory: ./q3_xgb_predictions_output
Please copy the generated part-*.csv to predictions.csv
```

在运行第三问程序的时候，我的程序一直产生如下报错：

```
root@b9890d3f18da:/data/assignment3# python Q3_3.py
WARNING: Using incubator modules: jdk.incubator.vector
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/05/08 22:46:56 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
26/05/08 22:46:56 WARN Utils: Service 'SparkUI' could not bind on port 4040. Attempting port 4041.
Traceback (most recent call last):
  File "/data/assignment3/Q3_3.py", line 192, in <module>
    LightGBMRegressor()
  File "/opt/spark-4.1.0-bin-hadoop3/python/pyspark/__init__.py", line 115, in wrapper
    return func(self, **kwargs)
  File "/opt/venv/lib/python3.12/site-packages/synapse/ml/lightgbm/LightGBMRegressor.py", line 406, in __init__
    self._java_obj = self._new_java_obj("com.microsoft.azure.synapse.ml.lightgbm.LightGBMRegressor", self.uid)
  File "/opt/spark-4.1.0-bin-hadoop3/python/pyspark/ml/util.py", line 396, in wrapped
    return f(java_class, *args)
  File "/opt/spark-4.1.0-bin-hadoop3/python/pyspark/ml/wrapper.py", line 104, in _new_java_obj
    return java_obj(*java_args)
TypeError: 'JavaPackage' object is not callable
```

经过研究作业给出的官方说明文档以及和大语言模型的对话，我发现SynapseML LightGBM 工具包的官方文档中，对于安装包的部分内容如下：

Version: 1.1.3

# Install SynapseML

## Microsoft Fabric

SynapseML is already installed in Microsoft Fabric notebooks. To change the version please place the following in the first cell of your notebook:

```
%%configure -f
{
  "name": "synapseml",
  "conf": {
    "spark.jars.packages": "com.microsoft.azure:synapseml_2.12:<THE_SYNAPSEML_VERSION_YOU_WANT>",
    "spark.jars.repositories": "https://mmlspark.azureedge.net/maven",
    "spark.jars.excludes": "org.scala-lang:scala-reflect,org.apache.spark:spark-tags_2.12,org.scalactic:scalactic",
    "spark.yarn.user.classpath.first": "true",
    "spark.sql.parquet.enableVectorizedReader": "false"
  }
}
```

## Synapse

SynapseML is already installed in Synapse Analytics notebooks. To change the version please place the following in the first cell of your notebook:

For Spark3.5 pools

而课程服务器上的spark版本是4.1，出现了版本不对应的问题，因此程序报错原因是 SynapseML 的 Java 侧类没有在 Spark 4.1 会话里正确加载，而且官方文档当前也没有 Spark 4.1 的安装方案，因此目前可能除了降低spark版本无法解决这个问题。